

An Aligned-Center Recursion for Products of q -Integers

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Abstract

Connelly, Ito, Martinez, Shevchenko, and Yang conjectured a sufficient condition for the unimodality of $\prod_{i=1}^k [a_i]_q [b]_{q^r}$, and proved necessity in two low-dimensional regimes. We prove their sufficient condition for every $k \geq 1$ and $r \geq 2$. The proof is an induction based on the identity

$$[a+r]_q [b+1]_{q^r} = q^r [a]_q [b]_{q^r} + [a+r(b+1)]_q.$$

After multiplication by the remaining factors, the two terms on the right are symmetric unimodal about exactly the same center. Allocating the available increments among the parameters completes the induction.

1 Introduction

For a positive integer n , write

$$[n]_q = 1 + q + \cdots + q^{n-1}.$$

Products of such polynomials arise naturally in questions about q -Fibonomial coefficients. Connelly, Ito, Martinez, Shevchenko, and Yang formulated the following sufficient condition [1, Conjecture 5.4]: if $r \geq 2$, $k \geq 1$, and $a_1, \dots, a_k, b$ are positive integers satisfying

$$r \mid a_i \quad \text{for some } i, \quad \text{or} \quad b \leq 1 + \sum_{i=1}^k \left\lfloor \frac{a_i}{r} \right\rfloor, \quad (1)$$

then

$$P(q) = \prod_{i=1}^k [a_i]_q [b]_{q^r} \quad (2)$$

is unimodal. They also conjecture necessity when $k \leq 3$ or $r \leq 3$, prove several precursor cases, and verify bounded instances.

We prove the universally quantified sufficient statement in (1). In particular, this settles its first corner beyond both low-dimensional regimes, $k = r = 4$, but the argument does not depend on either value.

Theorem 1. *Let $r \geq 2$, $k \geq 1$, and let $a_1, \dots, a_k, b$ be positive integers. If either condition in (1) holds, then the polynomial (2) is symmetric and unimodal.*

The condition is not necessary for general k, r . For example, the source observes that $([3]_q)^4 [2]_{q^4}$ is unimodal although it violates the inequality in (1) [1, after Conjecture 5.4]. Thus Theorem 1 proves only the sufficient direction and makes no new necessity claim.

2 Two elementary lemmas

A finite nonnegative coefficient sequence is *symmetric unimodal about C* if coefficients at exponents equidistant from C agree and the coefficients weakly increase up to C . We allow initial and terminal zeros; this convention makes translates convenient.

Lemma 2. *The product of two nonnegative symmetric unimodal polynomials is symmetric unimodal. A sum of nonnegative symmetric unimodal polynomials having a common center is symmetric unimodal.*

Proof. The assertion about sums follows coefficientwise. For products, decompose each symmetric unimodal coefficient sequence into a nonnegative linear combination of indicator sequences of nested intervals having the same center. The convolution of two interval indicators is a symmetric trapezoidal sequence, hence unimodal, and all resulting convolutions have the sum of the original centers. Summing them proves the claim. $\square$

The next identity supplies the induction.

Lemma 3 (aligned-center recursion). *For positive integers a, b, r ,*

$$[a + r]_q [b + 1]_{q^r} = q^r [a]_q [b]_{q^r} + [a + r(b + 1)]_q. \quad (3)$$

Proof. Use

$$[a + r]_q = [r]_q + q^r [a]_q, \quad [b + 1]_{q^r} = 1 + q^r [b]_{q^r}.$$

After subtracting the first term on the right side of (3), the remaining terms are

$$\begin{aligned} [r]_q [b + 1]_{q^r} + q^r [a]_q (1 + (q^r - 1) [b]_{q^r}) &= [r(b + 1)]_q + q^{r(b+1)} [a]_q \\ &= [a + r(b + 1)]_q. \end{aligned}$$

For a coefficient-level check independent of this manipulation, the left side counts pairs (x, j) with $0 \leq x < a + r$ and $0 \leq j \leq b$, weighted by $x + rj$. The pairs with $x < a$ and $j \geq 1$ give $q^r [a]_q [b]_{q^r}$. The complementary pairs have either $j = 0$, or $x \geq a$ and $j \geq 1$; their weights occur once each and fill the consecutive interval from 0 to $a + r(b + 1) - 1$. They give the other term in (3). $\square$

3 Proof of the theorem

Suppose first that $r \mid a_i$ for some i . Writing $a_i = rs$, we have

$$[a_i]_q [b]_{q^r} = [r]_q [s]_{q^r} [b]_{q^r}.$$

The product $[s]_z [b]_z$ is symmetric unimodal in z . Substitution $z = q^r$, followed by multiplication by $[r]_q$, repeats each of its coefficients in a block of length r ; the resulting sequence remains symmetric unimodal. Lemma 2 then permits multiplication by the remaining factors $[a_j]_q$. This proves the divisibility branch.

Now assume no a_i is divisible by r , and that the inequality in (1) holds. Put $c_i = \lfloor a_i/r \rfloor$. Choose integers d_i such that

$$0 \leq d_i \leq c_i, \quad \sum_{i=1}^k d_i = b - 1. \quad (4)$$

Such a choice exists by (1). Define $a_i^{(0)} = a_i - rd_i$. Every $a_i^{(0)}$ is positive because $r \nmid a_i$. At spacer length one, the base polynomial is

$$\prod_{i=1}^k [a_i^{(0)}]_q,$$

which is symmetric unimodal by Lemma 2.

Starting from this base, perform the $b - 1$ increments allocated in (4). At one step, let a be the current value of the selected length, let c be the current spacer length, and let $B(q)$ be the product of all other current ordinary q -integers. Lemma 3 gives

$$[a + r]_q B(q) [c + 1]_{q^r} = q^r [a]_q B(q) [c]_{q^r} + [a + r(c + 1)]_q B(q). \quad (5)$$

By induction, the first term on the right is a translate of a symmetric unimodal polynomial. The second term is a product of ordinary q -integers, hence is symmetric unimodal.

It remains only to check their centers. If the old degree is

$$E = (a - 1) + \deg B + r(c - 1),$$

then the new degree is $E + 2r$. The support endpoints of the translated first term sum to $E + 2r$, while the degree of the second term is

$$a + r(c + 1) - 1 + \deg B = E + 2r.$$

Thus both summands in (5) have center $(E + 2r)/2$. Their sum is symmetric unimodal by Lemma 2. Iterating (5) reconstructs all final lengths a_i and the final spacer length b , proving Theorem 1.

4 Reproducibility and scope

The identity and induction use no computation. The accompanying exact standard-library replay nevertheless checks (3) by both polynomial multiplication and the independent pair partition, checks center alignment at every recursive step, and compares the recursive construction with direct multiplication over a bounded regression corpus. From the archive root run

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python3 proof/qanalog_conjecture54_sufficiency.py
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The bounded corpus is regression evidence only; it is not an assumption in the universal proof.

The theorem proves the sufficient direction of Conjecture 5.4 for all k, r . It does not prove the conjectured necessity for $k \leq 3$ or $r \leq 3$, does not characterize the additional unimodal products outside (1), and does not by itself settle the general q -Fibonomial unimodality conjecture.

AI assistance. The author used AI assistance in developing the proof presentation and its verification code.

References

- [1] Brendan B. Connolly, Ezekiel Ito, Thomas C. Martinez, Olha Shevchenko, and Kacey Yang. Unimodality of q -fibonomial coefficients for small cases, 2026. arXiv:2605.12822v1, May 12, 2026.